



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 3 • STANDARD 6.AT.3A

Determine Equivalent Ratios Using Graphs

Lesson 3-4 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can graph the values from a ratio table as points on the coordinate plane.

Language Objective: I can explain my graph using the words coordinate plane, ordered pair, origin, and proportional.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Graph Ratio Tables

Coordinate plane

Ordered pair

Linear pattern

Proportional

Origin

Ratio table



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Each row of a ratio table becomes one
 on the coordinate plane.

2

The flat grid formed by a horizontal and a vertical number line is the
.

3

A set of two numbers (x, y) that names a point is an
.

4

When graphed points form a straight line, they show a
.

5 Two quantities that always keep the same ratio are .

6 The point (0, 0) where the two axes meet is the .

7 A table that lists equivalent ratios in rows or columns is a .

Write About the Math

A food truck tracks how many tacos it sells and how much salsa it uses. After recording data for a week, the owner plots the points on a graph and notices they form a straight line. This helps the owner predict exactly how much salsa to prepare for any number of tacos?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Graph Ratio Tables**

Representar tablas de razones en una gráfica

☐ **Coordinate plane**

Plano cartesiano

☐ **Ordered pair**

Par ordenado

☐ **Linear pattern**

Patrón lineal

☐ **Proportional**

Proporcional

Model: *The points form a straight line, and each sundae adds 2 ounces of sauce.* — Listen for 'straight line through the origin' and the connection that the constant 2-ounce step shows a proportional relationship.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- Every point on the line has the same ratio — ____ taco to ____ ounces of salsa — so 10 tacos would use ____ ounces.
- My answer is ____ because ____.

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.

Mi afirmación es ____.

- **My evidence is** ____, **so** ____.

Mi evidencia es ____, *entonces* ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Graph Ratio Tables
2. **Notes 2:** Coordinate plane
3. **Notes 3:** Ordered pair
4. **Notes 4:** Linear pattern
5. **Notes 5:** Proportional
6. **Notes 6:** Origin
7. **Notes 7:** Ratio table
8. **Watch for this mistake:** *A common mistake in Graph Ratio Tables is writing the ordered pair in the wrong order — putting the second table value first instead of matching (x, y) to the table's column order. For example, if a recipe table shows 2 cups of sugar for every 4 cups of flour, a student might plot $(4, 2)$ instead of $(2, 4)$, landing the point in a completely different spot on the grid. Before you plot, check: which quantity goes on the x -axis, and does your ordered pair start with that number?*
9. **Try It 1:** A. $(3, 6)$ — Cups (x) come first, scoops (y) second. 3 cups need $3 \times 2 = 6$ scoops, so the point is $(3, 6)$.
10. **Try It 2:** A. $y = 2x$ — Each cup needs 2 scoops, so y (scoops) is always 2 times x (cups): $y = 2x$. For 4 cups, $y = 2 \times 4 = 8$.